Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Darshan ease
Maximum Marks	1 Mark

Scalability & Future Plan:

The DarshanEase – Temple Appointment Booking System is designed with a scalable MERN Stack architecture that can support an increasing number of users, temples, bookings, and donations without major changes to the existing system. The modular design and RESTful APIs make it easy to add new features, improve performance, and integrate with additional services as the application grows.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Online payment gateway is not integrated.	Users cannot make online payments for donations or special darshan bookings.	**High**
2	Application is currently deployed only in a local environment.	Limits public access and scalability for real-time users.	**Medium**
3	Email and SMS notification features are not implemented.	Users do not receive booking confirmations or appointment reminders automatically.	**High**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	**Supports a limited number of concurrent users in the local environment.**	**Deploy the application on cloud platforms with load balancing and auto-scaling to support more users.**

2	Database Management	Uses MongoDB to store user, temple, booking, slot, and donation data.	Use MongoDB Atlas with database replication, sharding, and regular backups for improved scalability.
3	Performance	Provides stable performance under moderate user load.	Optimize APIs, implement caching, and improve database queries to enhance response time and efficiency.
4	Security	Uses JWT authentication and role-based access control.	Enhance security with HTTPS, OAuth integration, data encryption, and multi-factor authentication (MFA).

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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment using AWS/Azure and CI/CD Integration	2 Months	Improves application scalability, availability, and automated deployment.
Phase 2	Online Payment Gateway and Email/SMS Notification Integration	3–4 Months	Enables secure online payments and provides instant booking confirmations and reminders.
Phase 3	Mobile Application, QR Code Verification, and AI-Based Slot Recommendation	5–6 Months	Increases accessibility through mobile devices, simplifies temple entry with QR codes, and helps users choose less crowded darshan slots.